

LLM-ERP User Manual (English)

Version: v3.16 Conversational ERP (erpilot) **Audience:** First-time ERP users — Owner, Sales, Plant Manager, Purchaser, Warehouse **Reading time:** ~30 minutes to get productive; refer to §11 Troubleshooting when stuck **Languages:** System supports 繁體中文 and us English with instant switching

v3.13-v3.16 new features (must read)

-  **Settings page:** load demo / clear demo / upload business docs / configure AI API key (§3.5)
-  **CRM page:** Lead pipeline / Opportunity Kanban / Customer 360 with activity timeline (§3.6)
-  **AskAI floating button:** always-on "live AI coach" on every page (§3.7)
-  **Auto activity logging:** orders / lead conversions / opportunity stage changes auto-create CRM events

A Note for First-Time Readers

If you're not a "computer person" and the word "ERP" makes your head spin — **don't worry, we redesigned it for you.**

This LLM-ERP does not require you to memorize where menus live, what fields are called, or which button comes first. **If you can type, you can use it.** Open your browser, type into the chat box:

"List parts below safety stock" "Order 100 M6 bolts from Chang Jiang Precision, due next Friday"
"Change SO-2025-0042's delivery date to 6/10"

The AI assistant understands your intent and queries / creates / updates / cancels on your behalf. **Any action that writes data triggers a Confirm Card** that you must click — and if you click wrong, you have 90 seconds to undo.

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1. System Overview

1.1 What is LLM-ERP?

LLM-ERP is a **Conversational ERP** designed for **small manufacturers with 50–100 employees**.

"ERP" in plain English: **software that manages inventory, orders, purchasing, production, and reports** for a company. Traditional ERPs (SAP, Oracle) cost millions to license and require 1–3 months of staff training.

LLM-ERP flips that: you don't learn the interface; you **type what you want to do** and the AI does it.

1.2 Three Core Promises

Promise	Plain English
 Natural language operation	Just type what you want. No menu hunting.
 Confirm everything + 90-second undo	Every write (create/update/delete) shows a confirm card. Click wrong? You have 90 seconds to undo.
 Real-time, no waiting for month-end	Owner asks "how's today" — answer in 10 seconds, not a monthly report.

1.3 Who's it for?

This system is designed for **four roles in a 50–100-person manufacturer**:

Persona	Job	Device
 Owner (Mr. Wang)	Monitor company at a glance; check financials anytime	Office desktop, Chrome browser
 Sales (Steve)	Answer stock/price/delivery in front of customers, in seconds	Laptop (Chrome + VPN when traveling)
 Plant Manager (Lin)	Spot bottlenecks; release WOs; adjust scheduling	Office desktop + factory floor war-room screen
 Purchaser/Warehouse (Lina)	Create POs; receive goods; cycle counting	Desktop + USB barcode scanner

1.4 What can you actually do?

Four operation classes (industry term: **CRUD**):

Letter	Meaning	Example phrasing
C = Create	Add	"Order 100 M6 bolts from Chang Jiang Precision"
R = Read	Query	"List parts below safety stock"
U = Update	Change	"Change SO-2025-0042's due date to 6/10"
D = Delete	Cancel	"Cancel PO-2025-0107"

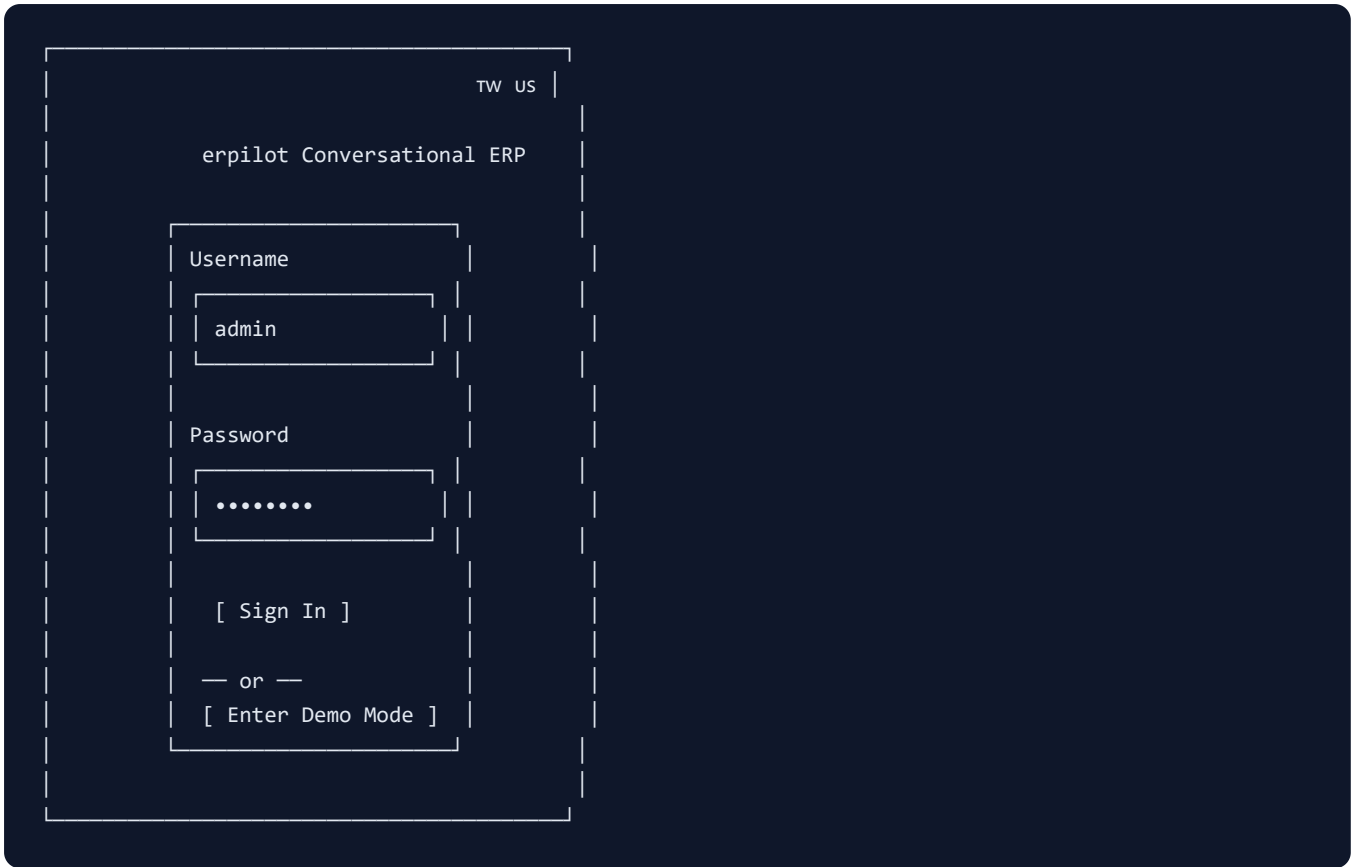
2. First Login

2.1 Open the system

After IT has deployed the system (if not, see [INSTALLATION_EN.md](#)), all you do is:

1. Open **Chrome** (latest version recommended; Edge / Firefox also work)
2. In the address bar, type: `http://localhost:5173` (or the URL IT gave you)
3. Press Enter

You'll see the login screen:



2.2 Enter credentials

Step	Action
1	Click the Username field
2	Type your username (default: admin)
3	Tab to or click the Password field
4	Type your password (default: admin123)
5	Click the blue [Sign In] button (or press Enter)

 **Change the default password right after first login!** [Top-right avatar](#) → [Profile](#) → [Change Password](#)

2.3 Don't want to memorize credentials? Demo mode

If you're just trying things out:

- Click **[Enter Demo Mode]**
- All features are unlocked for exploration

⚠ Demo mode is turned off in production by IT. Use real credentials when you go live.

2.4 Switch language

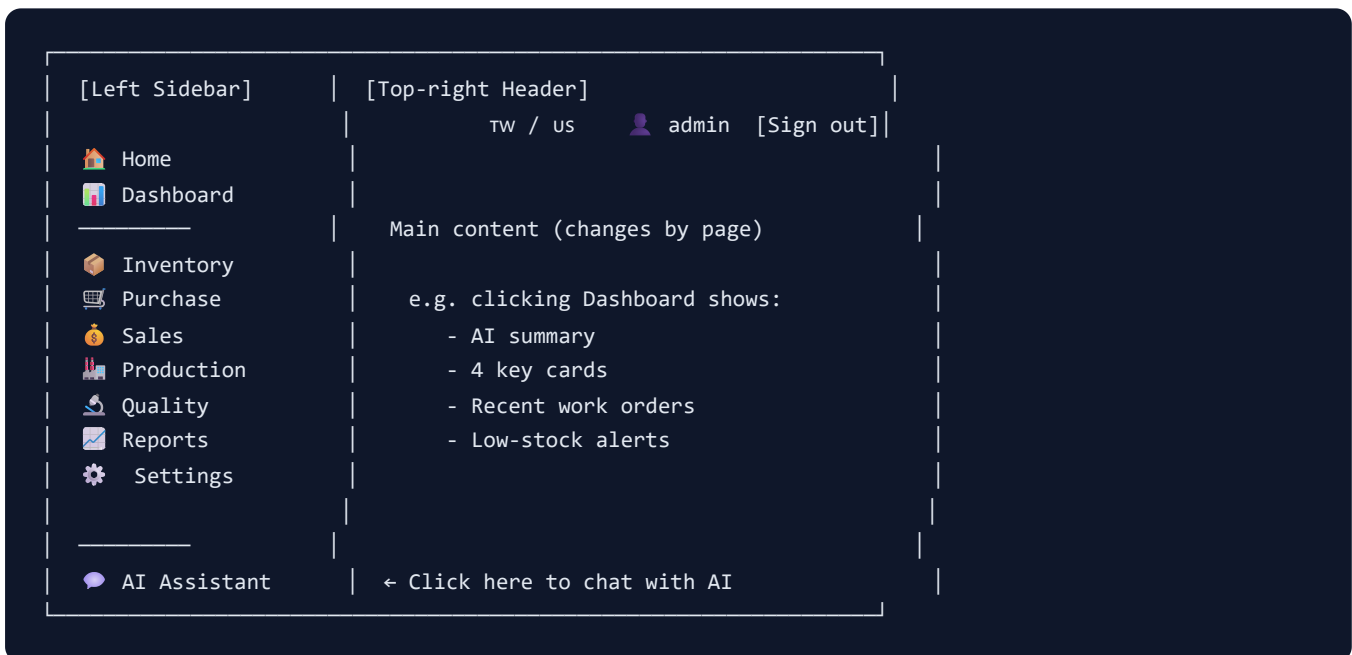
Top-right of the login screen has two flag buttons:

Flag	Meaning
tw	Switch to 繁體中文
us	Switch to English

Same toggle is in the top-right after login. Switching is instant — no re-login required.

3. Interface Tour

3.1 What the main screen looks like



3.2 Twelve pages in the left sidebar (v3.16+)

Three groups: Overview / Operations / System

Overview

#	Icon	Name	Purpose
1		Dashboard	KPI cards (revenue / WO / stock alerts) + charts
2		AI Assistant	Conversational CRUD: query/create/update/delete by talking
3		Event Stream	Real-time system events (new orders / WO complete / stock alerts)

Operations

#	Icon	Name	Purpose
4		Inventory	Parts, suppliers, stock queries
5		Purchase	PO creation, lookup, goods receipt
6		Production	Work orders (WO), progress tracking
7		Sales	SO, customer list
8		CRM ✨	NEW Lead pipeline / Opportunity Kanban / Customer 360 (see §3.6)
9		Quality	Inspection records (read-only audit ; intentional)

System

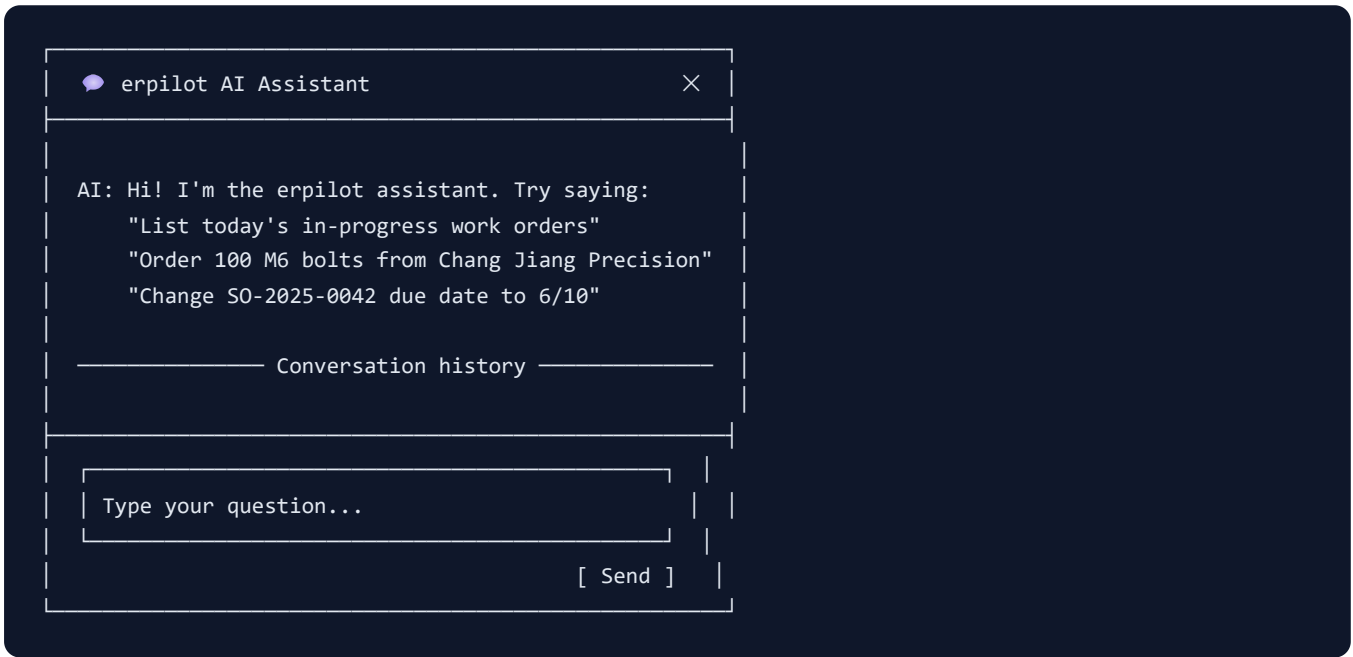
#	Icon	Name	Purpose
10		Permissions	Roles, permission assignment (admin only)
11		My Permissions	See your own permissions
12		Settings ✨	NEW AI config / load demo / upload files (see §3.5)

3.3 Three things in the top-right header

Element	Purpose
tw / us	Instant ZH/EN switch
 admin (your name)	Click for Profile / My Permissions / Change Password
[Sign out]	End your session; recommended before leaving for the day

3.4 Where's AI Assistant?

This is the most important v3.x feature. Click  AI Assistant in the sidebar; a chat panel opens:



How to use:

Step	Action
1	Click in the "Type your question..." box
2	Type, e.g.: "List today's in-progress work orders"
3	Press Enter (or click [Send])
4	Wait 2–10 seconds; the AI replies

3.5 Settings Page NEW in v3.13

Click  Settings in the sidebar. Four sections (everything a non-technical user needs is here):

A. 🤖 AI Assistant Configuration

First-time install without an LLM API key? **Paste it here yourself** — no command line needed.

Step	Action
1	Click the Provider dropdown → pick "DeepSeek (recommended)"
2	Click the "No account? Sign up →" link, follow steps for 5 min to get <code>sk-...</code>
3	Paste <code>sk-...</code> into the API Key field
4	Click 🧪 Test Connection (no save) — should show ✅ green
5	Click 💾 Save (effective immediately) — no restart needed

💡 Full guide: [HOW_TO_GET_LLM_API_KEY_EN.md](#)

B. 📦 Demo Data

Built-in: 5 customers / 3 suppliers / 10 parts (with `DEMO-` prefix). Clear after trial:

- ➕ **Load demo data** — writes demo data to DB (idempotent)
- 🗑️ **Clear demo data** — only removes `DEMO-` prefixed rows; your own data is safe

C. 📁 Upload Business Files

Upload customer quotes / invoices / specs / contracts (future AI parsing):

Step	Action
1	Pick category (quote / invoice / contract / spec / general)
2	Write a brief description (optional)
3	Drag file into the dropzone , or click to browse
4	Wait for ✅ upload success

Supports: PDF / Excel / CSV / Word / images, 25 MB max per file.

D. ⓘ System Info

Version, license track, commercial inquiry link.

3.6 🧡 CRM Page ✨ NEW in v3.15

Click 🧡 **CRM** in the sidebar. Three tabs:

Tab 1: 📋 Lead Pipeline

What's a Lead? A potential customer (from trade show / ad / referral) not yet a paying client.

4-column funnel:  New →  Contacted →  Qualified →  Lost (or  Converted)

Click  **Create Lead** to add. Qualified leads have a  **Convert to Customer** button.

Tab 2: 📁 Opportunity Kanban

What's an Opportunity? A customer with buying intent currently in active sales pursuit.

5 stages:  Prospect →  Proposal →  Negotiation →  Won →  Lost

Each card has a **Move to next** → button (no drag-and-drop needed). The top shows **Weighted Pipeline NT\$ X** = sum of (amount × probability%).

Tab 3: 👤 Customer 360

The killer feature. Pick one customer → see everything related in one view:

- Basic info (code / grade / contact)
- Stats (orders / opportunities / activities count)
- Orders list / Opportunities list
- **Activity timeline** — auto-populated when you place orders, convert leads, advance opportunities. Click  **Add Activity** for manual notes (calls / complaints / etc.).

3.7 💡 AskAI Floating Button ✨ NEW in v3.16 (erpilot original)

Purple 💡 button in the bottom-right corner of every page (except AI Assistant itself).

Purpose: Live AI coach for the page you're on. Stuck? Don't dig through docs — click and ask.

Features:

- Knows **which page you're on** (context-aware answers)
- 3 quick suggestion buttons: tap to send common questions
- No LLM key set? Shows a 3-step setup guide (not a cold error)

4. Talking to the AI: 4 CRUD Operations

This is the most important chapter — read it carefully.

Everything an ERP can do boils down to four classes: **Read / Create / Update / Delete**. We made all four typeable. Below: one or two examples per class.

4.1 Read — no Confirm Card, instant answer

Simplest case. AI answers directly (no data was changed).

Example 1: list parts below safety stock

You type:

List parts below safety stock

AI replies (about 3–5 seconds):

Found 3 parts below safety stock:

Part No	OnHand	Safety	Gap
M6-BOLT-20	300	500	-200
SUS304-PLATE	12	50	-38
GEAR-A-3MM	0	20	-20

Suggest restocking GEAR-A-3MM first (stockout).

Example 2: sales snapshot

You type:

What is this month's revenue? Top 3 customers?

AI replies:

This month (2026-05) revenue: NT\$ 4,820,000

Top 3 customers:

1. China Steel Structure NT\$ 1,800,000 (37%)
2. Delta Electronics NT\$ 980,000 (20%)
3. Hon Hai Precision NT\$ 720,000 (15%)



Read operations don't require a Confirm Card — the AI just answers.

4.2 Create — triggers ConfirmCard

Anything that writes data → AI always shows a Confirm Card.

Example 3: create a purchase order

You type:

Order 100 M6 bolts from Chang Jiang Precision, due next Friday

AI replies (no order created yet — a card appears):

ConfirmCard: please confirm this PO

Supplier	Chang Jiang Precision (S-0023)
Part No	M6-BOLT-20 (resolved from "M6 bolt")
Quantity	100
Unit price	NT\$ 0.5 (last contract price)
Total	NT\$ 50
Due date	2026-05-22 (next Friday)
Terms	Net 30 (supplier default)

[✓ Confirm]

[↶ Edit]

[✕ Cancel]

Auto-cancels in 90s ⌚ 88

Your action:

Want to	Click
Looks good, order it	[✓ Confirm]
Quantity wrong, fix it	[↶ Edit] (drops back to chat so you can say "make it 200")
Forget it	[✕ Cancel]

After clicking [✓ Confirm]:

✓ PO-2026-0142 created.
Notification sent to Chang Jiang Precision.
ETA: 2026-05-22

💡 Made a mistake? Within 90 seconds, say "undo last action" to revert.

4.3 Update — also triggers ConfirmCard

Changing data is just as risky as creating; the AI also shows a Confirm Card.

Example 4: change a sales order's due date

You type:

```
Change SO-2025-0042 due date to 6/10
```

AI replies:

 ConfirmCard: please confirm this update

Sales Order	SO-2025-0042
Customer	China Steel Structure (C-0007)
Item	GEAR-A-3MM × 500
Field	Due date
Old value	2026-06-03
New value	2026-06-10 ⚠ +7 days

[✓ Confirm] [↩ Edit] [✕ Cancel]

Click [✓ Confirm]:

✓ SO-2025-0042 due date updated to 2026-06-10.
Customer contact auto-notified by email.
Undo available for 90 seconds.

4.4 Delete / Cancel — red warning ConfirmCard

Delete is the most dangerous; the AI shows a red card with a strong warning.

Example 5: cancel a purchase order

You type:

```
Cancel PO-2026-0107
```

AI replies:

⚠

ConfirmCard: cancel PO (NOT recoverable)

PO

Supplier

Total

Status

PO-2026-0107

Da-Hua Precision

NT\$ 28,500

Sent, not yet received

📌

Impact:

- PO will be marked "Cancelled"

- Supplier will be notified (email)

- Incoming inventory reduced by 300

[✓ Yes, cancel]

[✗ No, keep the PO]

💡 **Note:** Quality inspection records cannot be deleted via AI — they are an audit trail and are kept forever.

5. ConfirmCard

5.1 Why ConfirmCard must exist

Because AI hallucinates (industry term for "makes things up").

For example:

- You say "Chang Jiang", AI might pick the wrong "Chang" supplier
- You say "100 units", AI might hear "1000 units"
- You say "M6 bolt", but inventory has M6-BOLT, M6-NUT, and M6-WASHER

Without ConfirmCard, AI executes immediately → **you spend 10× too much, or order from the wrong supplier.**

ConfirmCard design philosophy:

AI parses your sentence and drafts the action; a human gives the final sign-off by clicking confirm.

5.2 What's on a ConfirmCard

Every card has:

Section	Content
Title	What action this is (create PO / update SO / cancel WO ...)
Summary table	Every field, with the value parsed by AI
Impact note	What downstream effects this triggers
Three buttons	✓ Confirm / ✎ Edit / ✕ Cancel
Countdown	Auto-cancels after 90 seconds (in case you forget)

5.3 Four risk tiers behind the scenes

Tools are classified into four risk tiers; cards differ by color:

Tier	Color	Example	Card behavior
● Read	no card	"List inventory"	Instant answer, no card
● Soft-write	Yellow	"Save search filter"	Light card + easy undo
● Hard-write	Orange	"Create PO" / "Change due date"	Full ConfirmCard + 90s Undo
● Critical	Red	"Delete customer" / "Cancel large PO"	Red card + type-to-confirm

6. Slot-filling: When the AI Asks Back

6.1 What is slot-filling?

A "slot" is just a field. If you leave fields out, the AI **does not invent values** — it asks back.

6.2 Example: incomplete input

You type:

Send a PO to China Steel

AI replies:

OK, you want a PO to China Steel. Please tell me:

1. Which part? (Part No or description)
2. Quantity?
3. Due date?

You can answer in one line, e.g. "M6-BOLT-20, 1000 pcs, next Friday"

You reply:

M6 bolt 1000 pcs next Friday

AI replies (still ambiguous):

"M6 bolt" matches 3 parts – please pick one:

- | | |
|---|----------------------------------|
| ⓪ | M6-BOLT-20 (hex bolt, most used) |
| ① | M6-NUT (M6 nut) |
| ② | M6-WASHER (M6 washer) |

Reply with the number (e.g. "0") or the full Part No.

Reply  → AI shows the full ConfirmCard.

6.3 Why this design?

No follow-up questions = AI guesses = your company orders wrong. We'd rather ask a couple extra questions than cost you money.

 **Max 3 follow-ups:** if AI still can't pin it down after 3 questions, it suggests you switch to the [Purchase sidebar form](#) to fill it manually.

7. 90-second Undo

7.1 How to undo

Within 90 seconds of any create / update / delete, you can undo.

Method 1: type it

In the AI chat box:

Undo last action

or

Cancel the one I just did

AI replies:

✓

 Undid PO-2026-0142 creation.
PO marked as "Recalled".
Chang Jiang Precision was notified of the recall.

Method 2: click the inline button

After every write, the AI's response includes a small button:

✓

 PO-2026-0142 created.

[↩ Undo (67s left)]

Just click it.

7.2 Undo limitations

Situation	Can undo?
Within 90s, PO not yet received	✓ Yes
Within 90s, PO partially received	✗ No (downstream side effects)
Past 90s	✗ No — use the normal "Cancel document" workflow
Someone else confirmed it; you want to undo	✗ No — only the confirmer can undo

8. Four Personas in Action

8.1 🧑 Mr. Wang (Owner) — desktop morning briefing

Scenario: 8:30 AM, walks into the office, wants today's status.

Steps:

#	Action
1	Open Chrome → <code>http://localhost:5173</code>
2	Login as <code>boss_wang</code>
3	Lands on  Dashboard by default; AI summary appears at the top

AI summary looks like:

 Morning, Mr. Wang! Today's brief (2026-05-17)

 Yesterday: 12 shipments, NT\$ 580,000 revenue

 Pending releases today: 3 WOs (urgent SO-0042)

 Stock alert: M6 bolts 300 (< safety 500)

 AR DSO: 48 days (+3 vs last month)

[View details]

[Handle urgent]

[Notify Lina]

Want more? Click  AI Assistant and type:

What's this month's gross margin? Compared to last month?

AI replies:

This month gross margin: 23.4% (NT\$ 1,128K / NT\$ 4,820K)

Last month: 21.8%

Change: +1.6 pp 

Main driver: CNC machined parts margin rose from 18% to 22%.

8.2 Steve (Sales) — 3-second answer in front of customer

Scenario: Steve is at the customer's office. Customer asks: "Do you still have the GEAR-A I bought last time? How soon can you deliver?"

Steps:

#	Action
1	Open laptop Chrome (VPN already connected)
2	Login, click  AI Assistant
3	Type

Input:

```
GEAR-A-3MM for China Steel Structure – current stock and earliest commitable delivery?
```

AI replies (3–5 seconds):

```
China Steel Structure (C-0007) × GEAR-A-3MM:

On hand:           420
In production:    150 (W0-2026-0089, completes 5/22)
Reserved:         80
-----
Available to promise: 490
Earliest ship:      if ordered today, ship 5/23

🚩 Last price to China Steel: NT$ 4,500/pc (2026-04-20)
```

Customer says "Order 300":

```
Create SO for China Steel Structure: GEAR-A-3MM × 300, use last price, due 5/30
```

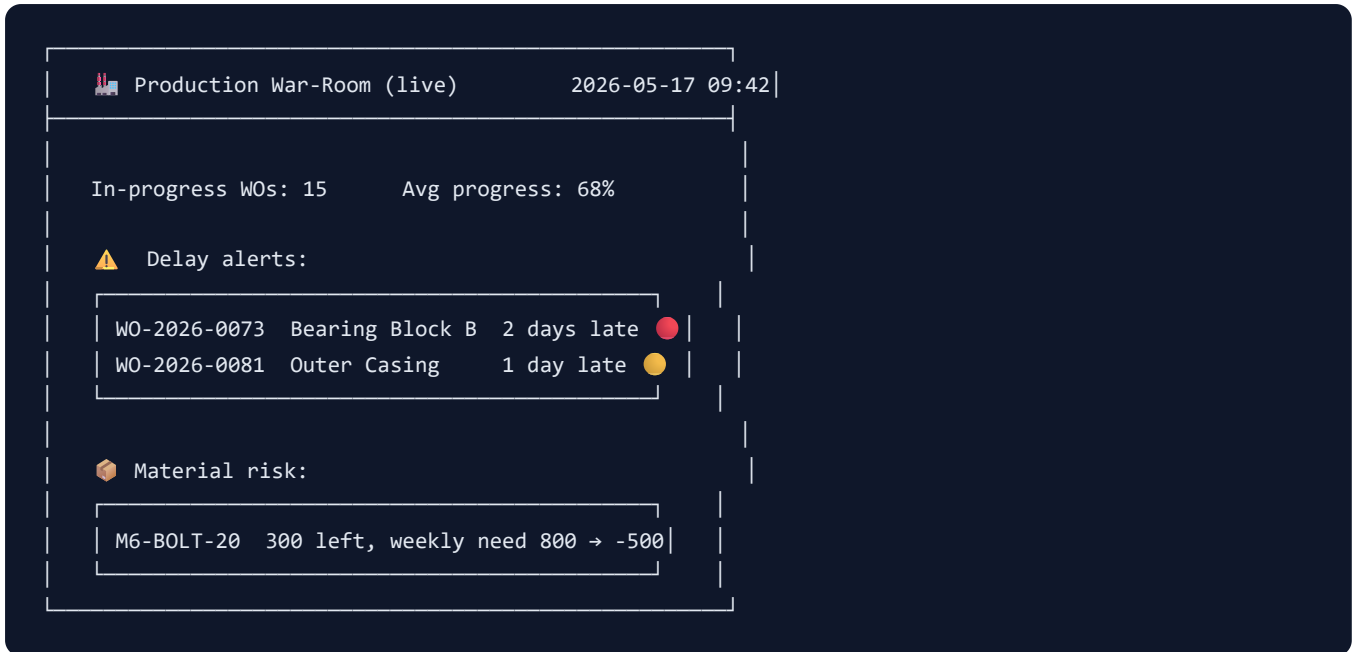
AI shows ConfirmCard, Steve clicks [✓ Confirm] → SO created in 30 seconds.

💡 Steve cannot see other reps' customers — this is RBAC (role-based access control).

8.3 Lin (Plant Manager) — desktop + war-room screen

Scenario A: desk view

Lin opens the dashboard, clicks  **Production**, and sees the war-room view:



Scenario B: release a WO via chat

Lin types:

```
Release WO-2026-0091 and assign to machine CNC-A
```

AI shows ConfirmCard, click [✓] → WO drops to the shop floor.

8.4 🧑🏻 Lina (Purchaser/Warehouse) — desktop + USB barcode scanner

Scenario A: PO via chat

Lina gets an email from Lin: "restock M6 bolts". She tells AI:

```
Order from Chang Jiang Precision: M6-BOLT-20 × 2000, due next Wed, Net 30
```

AI shows ConfirmCard, Lina confirms, PO created.

Scenario B: receive goods with USB scanner

#	Action
1	Open  Purchase → find <code>PO-2026-0142</code>
2	Click Receive — receiving form opens
3	Plug USB barcode scanner into the PC
4	Point at the supplier's box barcode, beep
5	System auto-fills Part No and expected quantity
6	Type the actual received quantity
7	Click Confirm Inwarding → stock +N, PO marked "Received"

 Advanced USB-scanner workflows (multi-bin, batch/lot tracking) are scheduled for **Phase 2**. The current release supports the basic "beep, get the Part No" workflow only.

9. Three Licensing Tracks

LLM-ERP ships under three licensing tracks — picking the right one can save millions.

Track	For whom	Cost	Highlight
 AGPL	Want to inspect source, OK with sharing your modifications	Free	Fully open-source
 Small Business	50–100-person factory, ≤ 20 concurrent users	Completely free	Includes closed-source connectors (Digiwin / Chain Sea integrations)
 Commercial	> 20 concurrent users, ISVs, SaaS providers	Negotiated	Removes AGPL clauses

How to apply for Small Business?

It's free, but registration is required. See:

-  `LICENSE-COMMERCIAL.md` (in the project root)
-  `docs/COMMERCIAL_LICENSING_FAQ_ZH.md`

Email the licensing contact with your company name, headcount, and expected concurrent users. You'll receive a Small Business grant letter.

10. FAQ

Q1: Does the AI handle typos?

→ Mostly yes. "M6 bolt", "m6 buoy", "M6 螺絲" all attempt to resolve. When unsure, the AI asks back.

Q2: What if the AI answers wrong?

→ For Read: re-ask with clearer phrasing. For Create/Update/Delete: if the ConfirmCard looks wrong, click [➡ Edit] or [✕ Cancel] — do not confirm.

Q3: How long is chat history kept?

→ Default 30 days (IT-configurable). To view: top-right  avatar → "My Conversations".

Q4: Can I switch roles, e.g., a salesperson viewing the plant manager's screen?

→ One account = one role; you can't self-switch. Ask IT / the owner to add a role under  Settings → Users.

Q5: Can I export to Excel?

→ Every list page has  **Export CSV** in the top-right (CSV opens in Excel). Native  export is scheduled for Phase 2.

Q6: Forgot my password?

→ Ask IT to reset. Self-service email reset is scheduled for Phase 2.

Q7: Can I use voice input?

→ Phase 4 (Whisper speech-to-text). Currently text only.

Q8: Can multiple factories share one system?

→ Yes — it's called **MESH multi-factory**: each site's data stays local, HQ runs aggregate queries. See  **NETWORK_DEPLOYMENT_EN.md**.

Q9: AI not responding?

→ See §11.3 Troubleshooting.

Q10: What do I need before commercial use?

→ See Chapter 9 plus `LICENSE-COMMERCIAL.md`.

11. Troubleshooting

11.1 Can't reach the login page

Symptom	Fix
Browser says "Cannot reach this site"	Have IT confirm backend is up: <code>docker compose ps</code>
503 / 504 errors	Ask IT to restart: <code>docker compose restart backend</code>
401 keeps popping up	Press F12 → Application → Storage → Clear site data, refresh

11.2 No data shown

Symptom	Fix
Empty list	IT hasn't seeded demo data (<code>docker compose exec backend python -m scripts.seed</code>)
Only see customers you created	By design (RBAC). Not a bug
Button returns 403	Your role lacks permission — ask IT to grant

11.3 AI Assistant unresponsive

Symptom	Fix
Shows "[demo mode]"	IT hasn't set the LLM API key in <code>.env</code> (<code>LLM_API_KEY</code>)
Shows "LLM call failed"	Key may be expired / out of quota — ask IT
Long wait (> 30s)	LLM is thinking; allow up to 60s; otherwise check backend logs
AI keeps asking the same thing	After 3 follow-ups it will suggest using the sidebar form instead

11.4 ConfirmCard issues

Symptom	Fix
Card auto-cancelled (countdown hit zero)	Just re-ask the AI
[ Edit] does nothing	Refresh (F5) and re-ask
[ Confirm] spins forever	Look for error in the bottom-right; have IT check the backend

11.5 Undo failed

Symptom	Fix
"Past 90 seconds, cannot undo"	Use the formal "Cancel document" workflow
"Action consumed by downstream flow"	E.g., PO partially received — return the goods first
"You are not the executor"	The person who confirmed must undo

11.6 Still stuck?

Contact internal IT and include:

- Screenshot (Win+Shift+S)
- The exact phrasing you typed
- Error message (if any)
- Timestamp

Or email your company's ERP support contact.

Manual version: v3.16 (2026-05-17) — includes Settings page / CRM page / AskAI floating button / Auto CrmEvent Chinese version: [USER_MANUAL_ZH.md](#) Companion docs: [CONVERSATIONAL_ERP_DESIGN_EN.md](#) (architecture) / [LICENSE-COMMERCIAL.md](#) (licensing)